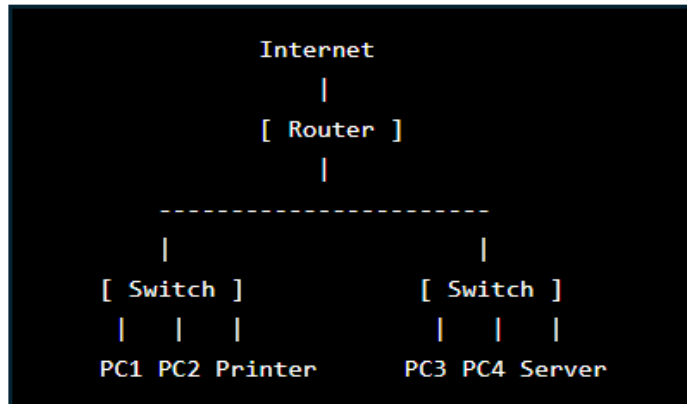


Feature	Router	Switch
Main Function	Connects different networks together, such as a home network and the Internet.	Connects multiple devices within the same Local Area Network (LAN) .
Address Used	Uses IP (Internet Protocol) addresses to send data between different networks.	Uses MAC (Media Access Control) addresses to send data to the correct device within the same network.
Network Layer	Works at Layer 3 (Network Layer) of the OSI model.	Works at Layer 2 (Data Link Layer) of the OSI model (some advanced Layer 3 switches can also route).
Traffic Handling	Routes data packets between different networks by choosing the best available path.	Forwards data frames only to the destination device inside the LAN, reducing unnecessary traffic.
Broadcast Domains	Separates broadcast domains, reducing network congestion.	Does not separate broadcast domains (unless using VLANs), but creates separate collision domains for each port.
Speed	Usually slower than a switch because it analyzes IP addresses and routing information.	Faster because it only checks MAC addresses and forwards frames directly.
Security	Provides features like firewalls, NAT (Network Address Translation), access control lists (ACLs), and routing policies .	Provides basic security features such as port security, MAC filtering, and VLAN support .
Example	A Wi-Fi router connecting your home network to the Internet.	A switch connecting computers, printers, and servers inside an office.

Real-Life Example

Imagine a company network:



- The **router** connects the company network to the Internet.
- The **switches** connect the computers, printer, and server within the company's LAN.

Key Differences

Router	Switch
Connects different networks	Connects devices within the same LAN
Uses IP addresses	Uses MAC addresses
Works at Layer 3	Works at Layer 2
Routes packets between networks	Forwards frames within a network
Uses routing tables	Uses a MAC address table

Router

Provides firewall, NAT, and routing features

Slightly slower due to routing decisions

Switch

Provides fast switching and VLAN support

Faster for local communication